Week 2 print and bring checklist

Delivery to a target. 12 children, four teams of three, two robots. Finish the opening explanation by minute 14.

Teaching files

- Slides with embedded notes and slide PDF.

- Instructor notes, Word notes and printable notes.

- Full Python program. The slide shows only the student recipe.

- Cargo-tray activity and build PDF.

Print

- [ ] **One copy of the 14-page class packet, single-sided**: 5 team sheets (4 + 1 spare), 4 cargo-tray build cards, 4 wheel reference cards and 1 attendance sheet. Do not also print the individual files.

- [ ] Optional: one copy of this two-page checklist. Packet plus checklist = **16 sheets**.

- [ ] Reuse role cards and hub guides. If missing, print two copies of both pages of Role_Cards.pdf and two copies of the hub guide. That adds 6 sheets. Cut role cards into 12 cards.

- [ ] Read notes digitally or print one lead copy. Do not print the workbook or slides for every child.

Pack

- [ ] 2 charged assembled SPIKE bases/hubs labeled A and B. Left motor A, right B, auxiliary unplugged. No sensors needed.

- [ ] 4 laptops and 4 chargers, 2 tested USB data cables + 1 spare, hub charging supplies. Save files offline and enable calculator access.

- [ ] 4 labeled team trays with equal build packs, 4 mounting/clearance templates, 2 tested mentor cargo trays and 4 storage bags. Keep chassis pieces out of build packs.

- [ ] 14 pencils, 14 erasers, 2 sharpeners, 4 rulers, 2 measuring tapes, 4 clipboards and 1 records folder.

- [ ] 2 rolls floor-safe tape, 2 rolls craft tape, 2 pairs child-safe scissors, 2 timers, 4 station labels and the instructor display/projector setup.

- [ ] 6 pre-cut card sheets for trays (4 teams + 2 mentor trays), or the prepared LEGO beams/pins in the build packs.

- [ ] 6 similar lightweight cubes (4 build-test cubes + 1 per robot).

- [ ] 6 non-stretch strings or paper strips about 40 cm long (4 teams + 2 mentor sets).

- [ ] 2 floor lanes at least 100 cm long with extra stopping space as rehearsal requires, 2 targets at 40 cm and 2 fixed chassis-reference markers.

- [ ] Optional: 1 loose matching wheel already available, or 1 prepared paper/card circle. Do not dismantle a teaching base for the demonstration.

Rehearse

- [ ] With power off, measure each tire's diameter and circumference C. Confirm direct wheel-to-motor attachment. Write each kit's C on its team sheets. The 5.6 cm / 17.6 cm slide example is not a measurement of these kits.

- [ ] Practice the two-minute motor explanation: electricity in a coil creates magnetism, magnetic forces turn the shaft, and tire grip moves the robot. The picture shows a simplified motor; the hub controls the real motors.

- [ ] Calculate predicted cm = (wheel degrees / 360) × C and target degrees = (40 / C) × 360. Round only the final target answer. Use cm for both target and C.

- [ ] On BOTH kits, rehearse 180, 360 and the calculated 40 cm value at SPEED = 200. Keep the 0–1080 degree limit. Check intended hub, A/B ports, stop controls and saving.

- [ ] Measure movement of the same chassis mark. Keep tray, cube, floor, heading and speed fixed. If the target calculation exceeds 1080, check measurements; do not bypass the guard. A 1 meter example is a paper calculation only today.

- [ ] Inspect both mentor trays and rehearse a quick team-tray swap. Test a different tray separately using the same code.

- [ ] Open the 11-slide deck, matching notes, full program and build references offline. Practice slides 1–8 within 14 minutes; coach calculator work at stations as needed.

- [ ] Assign two robot mentors, two build mentors, an adult lead and any floaters. Confirm local supervision and pickup arrangements.

Rotation and closeout

14–26: T1/T3 robots, T2/T4 build. 26–38: switch. 38–44: T1/T3 retest. 44–50: T2/T4 retest. Rotate jobs every 4 minutes in first turns and every 2 in retests. Reflect 50–54; save/reset 54–60.

- [ ] Save four projects and four labeled trays. Collect sheets, count parts, stop hubs and assign charging.

- [ ] Record actual measurements, rehearsal outcomes and shortages in SESSION_PREP.md. Keep child records in the approved private system.

Kit A: diameter ____ cm; C ____ cm; target ____ degrees; rehearsal/date _____

Kit B: diameter ____ cm; C ____ cm; target ____ degrees; rehearsal/date _____